Tutorial 9 Solution

Q1 (a) No, it is not a partition
(b) Yes, it is a partition
(c) Yes, it is a partition.
(d) No, it is not a partition.

Q2. (a) It is a poset
(b) Not it is not a poset since it is not antisymmetric
(c) It is a poset
(d) poset.

Q3. (a) No { is anyone taller than himself - No }.
(b) No: This is not a poset because it is not antisymmetric. Consider that we have a person a and person b and $a \neq b$ then the order pairs (a,b) can exist in the relation because we can have a & b be the same height

(c) This is a poset.

(d) This is not a poset because it is not antisymmetric Consider that you have two friends, a & b then the ordered pairs (a,b) & (b,a) satisfy the relation.

Q4. (a) This is a poset. The only ordered pairs we will have in this relation is (a,a) for all $a \in \mathbb{Z}$. This would mean that the relation is Reflexive, antisymmetric and transitive.

(b) This is not a poset because it is not Reflexive. It is also not antisymmetric

(c) This is a poset.

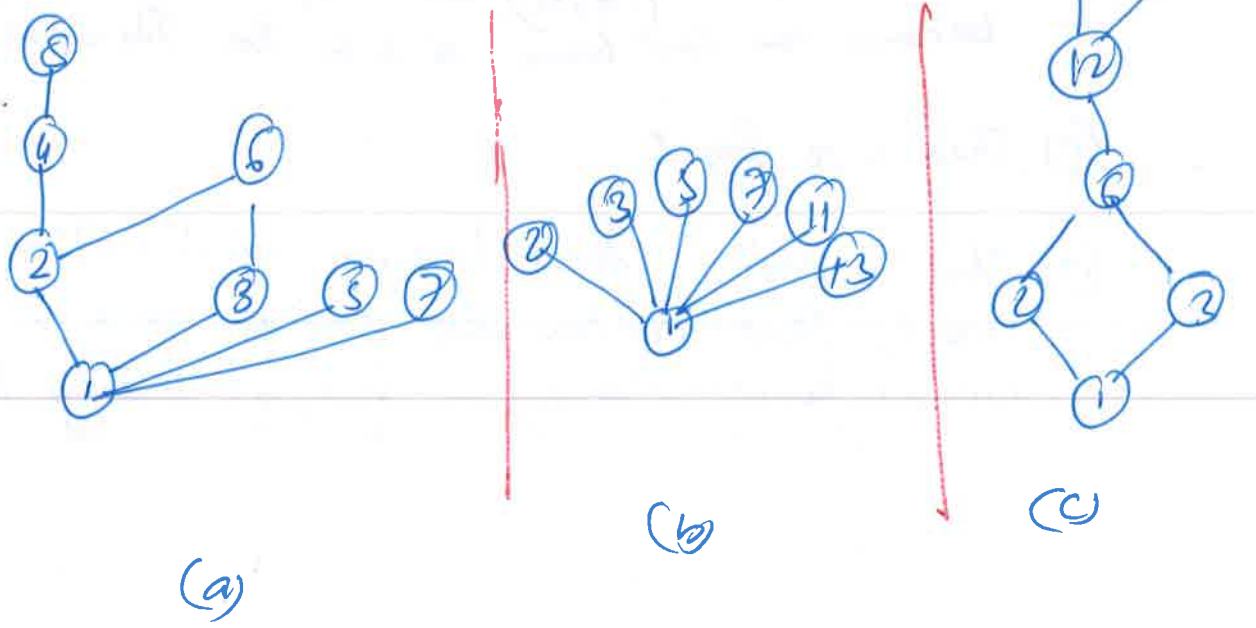
(d) This is not a poset because it is not Reflexive. It is also not antisymmetric

Q5. (a) No (b) Yes. (c) No

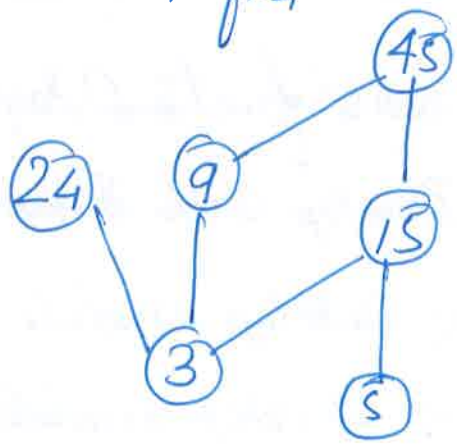
Q6. (a) $\{0\}$ & $\{1\}$

(b) 4 & 6.

Q7.



Q8 we will first draw the Hasse diagram



(a) maximal element = 24 and 45

(b) minimal element = 3 and 5

(c) There is No greatest element because this element would have to be a number that all other elements divide. Since our maximal elements are 24 and 45 and they do not divide each other, we do not have a greatest element

(d) There is no least element.

(e) 15 & 45

(f) least upper bound = {15}.

(g) 3, 5 and 15

(h) 15

Q9. (a) Yes, Every two elements will have a least upper bound and greatest lower bound.

(b) No, if we take elements b & c, then we will have f, g & h as the upper bound, but none of them will be the least upper bound.

Q10

Determine user needs $\prec$ write functional Requirements $\prec$ Set up test sites $\prec$ Develop such Requirements $\prec$ write documentation $\prec$ Develop module A $\prec$ Develop module B $\prec$ Develop module C $\prec$ Integrate module $\prec$ α test $\prec$ β test $\prec$ Completion.

(use Topological Sort)